

Independent Adversarial Audit Record

ERDOZ81 — Paper I — Finite Fractional Split-Graph Theorem

Primary target: **ERDOZ81-P1-FRAC-THM-01** (bundle: step23_frac_audit_ready)

Audited claim. For every split graph G on n vertices, $|E(G)| - 2v_3^*(G) \leq n^2/6 + n$, where $v_3^*(G)$ is the maximum fractional triangle packing.

Auditor stance. Fully independent and adversarial. The chain was reconstructed from scratch. The bundle embeds prior verdicts, a prior audit PDF, superseded proofs, and the auditor's own earlier scripts; none of these were used as authority. Every load-bearing identity, inequality, LP-duality step, and boundary case was re-derived by hand and re-checked by independent computation (exact algebra + a brute-force fractional-packing LP over 3,000 random split graphs that does not use any construction from the bundle).

1. Package-Completeness Result

Result: QUALIFIED — archive verified, but the completeness gate could not be executed as specified.

- **Archive integrity: PASS.** The supplied SHA-256 sidecar matches the archive exactly (4b7aa3e8...edb23a20). The separately re-uploaded copy is byte-identical to the first, confirming the correct file was audited.
- **Missing governing files.** The instruction directs the auditor to start from `README_FIRST.md` and verify all mandatory files against `PACKAGE_MANIFEST.md`. Neither file exists in this bundle, and there is no per-file hash manifest. File-level integrity therefore could not be cryptographically verified; assurance rests only on the archive-level checksum. All mathematical artifacts needed to audit the claim are nonetheless present and readable.
- **Clean-context independence compromised.** Unlike the clean v2 bundle, this package re-introduces exploratory history, prior gate verdicts, superseded proofs, the prior `INDEPENDENT_AUDIT_RECORD` PDF, and the auditor's own prior computation scripts. Per protocol these are excluded from a clean audit; they were disregarded as authority here.

2. Independence Disclosure

The audit relied solely on the mathematical statements in the supplied proofs, re-derived independently. It did **not** rely on: the embedded prior verdict (`RECOMMEND_PASS` on `BOUND-01`), the prior audit PDF/evidence record, the bundled `verify.py/exact.py/symb.py/build_pdf.py`, the gate records' self-audit/self-assigned statuses, or any superseded/archived material. Independent tooling written fresh for this audit: hand re-derivation; and a from-scratch fractional-packing LP harness over random split graphs (SciPy/HiGHS), plus exact-fraction spot checks.

3. Claim-by-Claim Findings (Load-Bearing Chain)

The seven original T1/T2/T3 gates underpinning $M(\kappa)$ (definitions, concavity, profile reduction, symmetrization, orbit-program equivalence, exact orbit solution, $q=0$) are **mathematically identical** to the clean v2 bundle already verified three independent ways; a byte-diff shows the only changes in the two altered proofs are cosmetic wording ("target" → "historical", "identity" → "corrected identity") with no mathematical change. They were re-confirmed and are treated as VERIFIED. Findings below focus on the genuinely new fractional-assembly gates.

3.1 Aggregate-load identity (LOAD-01)

$\sum_{e \in E(K)} \kappa_e = \frac{1}{2} \sum_{v \in V} d_v = b_{\geq 2}/2$. Verified by exchanging summation order: each v contributes $C(d_v, 2)/(d_v - 1) = d_v/2$. Edge count $|E(G)| = C(p, 2) + b_{\geq 2} + b_1$ follows from independence of I . Confirmed on all 3,000 test graphs. **VERIFIED.**

3.2 Phase-I feasibility and value (PHASE1-01)

With $\lambda_e = \min\{1, 1/\kappa_e\}$ and $w_1(vxy) = \lambda_{xy}/(d_v - 1)$: clique-edge load = $\min\{\kappa_e, 1\} \leq 1$; cross-edge load $\leq (d_v - 1) \cdot 1/(d_v - 1) = 1$ (feasible); value $v_1 = |H| + \sum_L \kappa_e$. Each positive triangle carries exactly one clique edge, so the value sum is exact. **VERIFIED.**

3.3 Finite strong LP duality (LP-DUALITY-01, Farkas)

Self-contained proof of $v = \tau$ for the edge-triangle packing/covering LP: weak duality is the standard swap; the cone is shown closed via a minimal-representation + linear-independence argument; separation is by nearest-point projection; strong duality follows from a Farkas certificate above each $a > v$ plus compactification of capped dual solutions. The degenerate no-triangle case is handled. The argument is correct and imports no external theorem. **VERIFIED**.

3.4 Phase-II dual reduction (PHASE2-DUAL-01)

Residual capacities $r_e = \max\{1 - \kappa_e, 0\}$; dual capped to $0 \leq x_e \leq 1$ (valid: capping preserves each triangle sum ≥ 1 and cannot raise the objective since $r_e \geq 0$); substitution $y = 1 - x$ turns $\sum_{e \in T} x_e \geq 1$ into $\sum_{e \in T} y_e \leq 2$; the signed extension over hot edges is legitimate because the coefficient $1 - \kappa_e \leq 0$ on H , so zeroing H coordinates preserves feasibility and the maximum; $z = 1 - y$ lands in P_p , yielding $v_{\text{con}} = v_1 + v_2 = b_{\geq 2}/2 + M(\kappa)$. Every step re-derived and checked. **VERIFIED**.

3.5 Exact deficit identity — the previously-open interface

Combining the edge count with $v_{\text{con}} = b_{\geq 2}/2 + M(\kappa)$ gives, with the $b_{\geq 2}$ terms cancelling exactly, $|E(G)| - 2v_{\text{con}} = C(p, 2) - 2M(\kappa) + b_1$. This is the identity the *prior* audit flagged as merely asserted; it is now **constructively proven** via the explicit combined packing $w_{\text{com}} = w_1 + w_2$ (feasible: clique-edge load $\min\{\kappa, 1\} + \max\{1 - \kappa, 0\} = 1$) plus the internal strong duality. Confirmed exactly on all 3,000 graphs. **VERIFIED — gap closed**.

3.6 Quantitative bound insertion (BOUND-01)

$M(\kappa) \geq R(p, q) - p/2$ with $R = (2p^2 - 2pq - q^2)/12$ — unchanged from the previously fully-verified gate; independently re-confirmed. Then $C(p, 2) - 2R + p = (p+q)^2/6 + p/2$ (exact algebra), so $|E(G)| - 2v_{\text{con}} \leq (p+q)^2/6 + p/2 + b_1$. **VERIFIED**.

3.7 Conversion to n and closure

$n = p + q + b_1 + |I_0|$ gives $(p+q)^2 \leq n^2$ and $p/2 + b_1 \leq p + b_1 \leq n$. Feasibility of w_{com} gives $v_{\text{con}} \leq v_3^*(G)$, hence $|E(G)| - 2v_3^*(G) \leq |E(G)| - 2v_{\text{con}} \leq n^2/6 + n$. The loss ledger ($p/2$ once, b_1 once, combined $\leq n$) is complete; isolated vertices enter only through n . **VERIFIED**.

4. Independent Derivations and Falsification Attempts

(a) Full-theorem LP test. 3,000 random split graphs (p up to 7, arbitrary independent-vertex neighborhoods incl. degree-0/1/2). For each, $v_3^*(G)$ was computed by an independent max fractional-packing LP over *all* triangles of G . The theorem held every time; **zero failures**.

(b) Every intermediate identity. Edge count, load identity, $M(\kappa) \geq R - p/2$, $v_{\text{con}} = b_{\geq 2}/2 + M(\kappa)$, the deficit identity, and feasibility $v_{\text{con}} \leq v_3^*$ were checked on all 3,000 graphs — all exact, zero failures.

(c) Tightness / masking check. The only equality case in the sample is the trivial empty graph ($n=0$). All non-trivial graphs carry strictly positive slack, so no near-violation is masked by tolerance. The complete-graph family $K_{2..8}$ confirms comfortable margins (e.g. K_8 : 9.33 vs 18.67).

(d) Duality re-derivation. Weak duality, capping validity, the signed-extension step, and the $y \leftrightarrow z$ change of variables were each re-derived by hand and matched the proofs. The Farkas construction was checked step-by-step (cone closedness, projection separation, certificate normalization $\lambda > 0$).

(e) Adversarial probes. Degenerate regimes were targeted: all-hot edges ($H=E(K)$, $L=\emptyset$), $q=0$, $p < 3$ (no clique triangles), and isolated/degree-1 vertices. The identities and the bound held in every case, including the degenerate $v_{\text{con}} = C(p, 2)$ closure.

5. Dependency Table

Dependency / Interface	Status
Archive checksum vs supplied .sha256	VERIFIED

Dependency / Interface	Status
README_FIRST.md / PACKAGE_MANIFEST.md present; file-level hash manifest	FAILED (absent)
DEF-01 well-definedness (v_3^* , P_p , M exist)	VERIFIED
LOAD-01 aggregate-load identity + edge count	VERIFIED
PHASE1-01 Phase-I feasibility and value	VERIFIED
LP-DUALITY-01 internal Farkas strong duality	VERIFIED
PHASE2-DUAL-01 dual reduction / capping / signed extension	VERIFIED
Exact deficit identity $ E -2v_{con}=C(p,2)-2M+b_1$	VERIFIED
BOUND-01 chain $M(\kappa) \geq R-p/2$ (7 gates, math unchanged)	VERIFIED
FRAC-THM-01 n-conversion and final closure	VERIFIED
Feasibility $v_{con} \leq v_3^*(G)$	VERIFIED
Fractional→integral transfer (declared future stage)	UNCHECKED (out of scope)
General (non-split) graphs	UNCHECKED (out of scope)

6. Material Objections, Minor Observations, Required Actions

Material objections (mathematics): None. No incorrect identity, missing regime, reversed inequality, unstated hypothesis, non-exhaustive optimization, or dependency mismatch was found. The fractional theorem and its full load-bearing chain are correct and independently confirmed.

Minor observations: (i) DEF-C1 contains a markdown typo (“inom p2” for the binomial) — cosmetic. (ii) Gate records still carry self-assigned PROVISIONAL_PASS / CANDIDATE / UNVERIFIED metadata; this audit supplies the missing independent verification of the mathematics. (iii) The theorem is presentation-relative in intermediate steps (p, q depend on the chosen split presentation) but the final n -bound is presentation-independent, so this is not a defect.

Required actions (packaging / governance, not mathematics): (1) Ship a README_FIRST.md and a per-file PACKAGE_MANIFEST.md with SHA-256 hashes so the completeness gate can be executed as specified. (2) For a strictly clean audit, exclude prior verdicts, prior audit artifacts, and auditor scripts. (3) Treat the fractional→integral transfer and any general-graph extension as separate gates requiring their own audit.

7. Final Verdict

RECOMMEND_PASS — on ERDOZ81-P1-FRAC-THM-01 (mathematics)

Every load-bearing link of the finite fractional split-graph theorem — well-definedness, the aggregate-load identity, Phase-I feasibility, internally-proved strong LP duality, the Phase-II dual reduction, the exact deficit identity, the quantitative bound $M(\kappa) \geq R-p/2$, and the conversion to n — was independently verified without material objection, by hand and by an independent fractional-packing LP over 3,000 split graphs (zero failures). Notably, the exact deficit identity that the prior BOUND-01 audit could only flag as asserted is now constructively established, closing that interface.

This PASS is qualified by one non-mathematical caveat: a strict reading of the package-completeness gate returns PACKAGE_INCOMPLETE because README_FIRST.md and PACKAGE_MANIFEST.md are absent, so file-level integrity is assured only by the matching archive checksum, and the clean-context independence conditions are violated by embedded prior verdicts and scripts (which were not relied upon). The mathematical verdict is unaffected, but a fully protocol-compliant certification requires the governing files above. Scope: this record certifies only the finite fractional theorem for split graphs; it does not certify the fractional→integral transfer or any general-graph result.